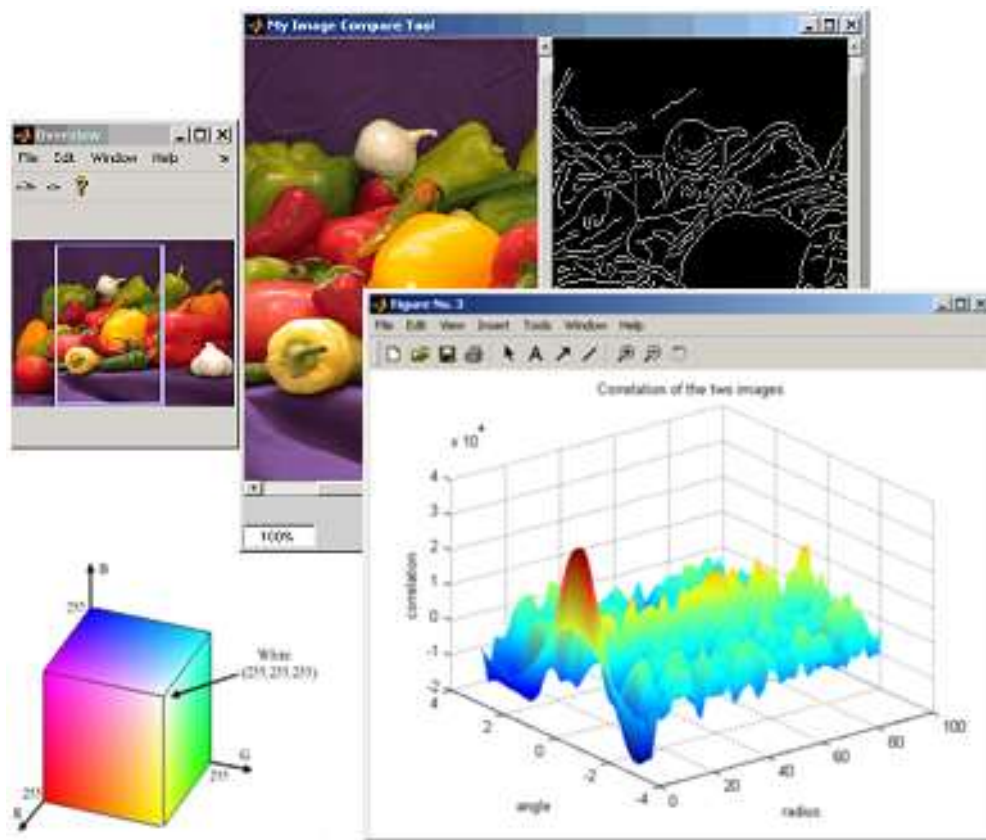


Digital Image Processing

LECTURE 10: SPATIAL FILTERING



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Spatial Filtering

Image Enhancement

❑ Classification of Image Enhancement Techniques:

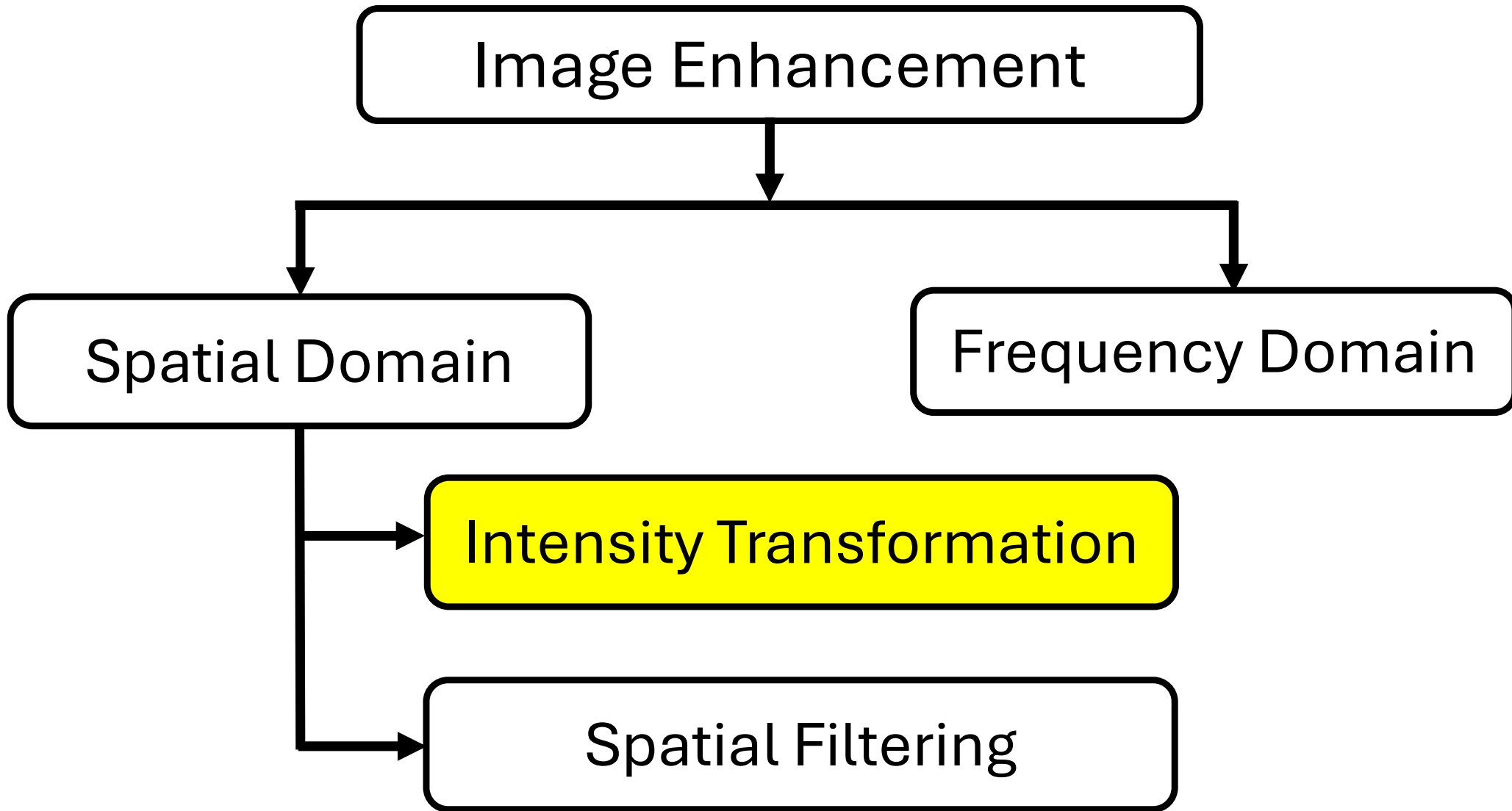


Image Enhancement

□ Spatial Domain:

- The term spatial domain refers to the image plane itself, and image processing methods in this category are based on direct manipulation of pixels in an image.

- Two principal categories:

- **Intensity Transformation:** Intensity Transformation operates on single pixels of an image for tasks.

Examples: Contrast Stretching, and Image thresholding

- **Spatial Filtering:** Spatial filtering performs operations on the neighbourhood of every pixel in an image.

Examples: Image smoothing, image sharpening, Image averaging

Spatial Filtering

Mask

3×3

m1	m2	M3
m4	m5	M6
m7	m8	m9

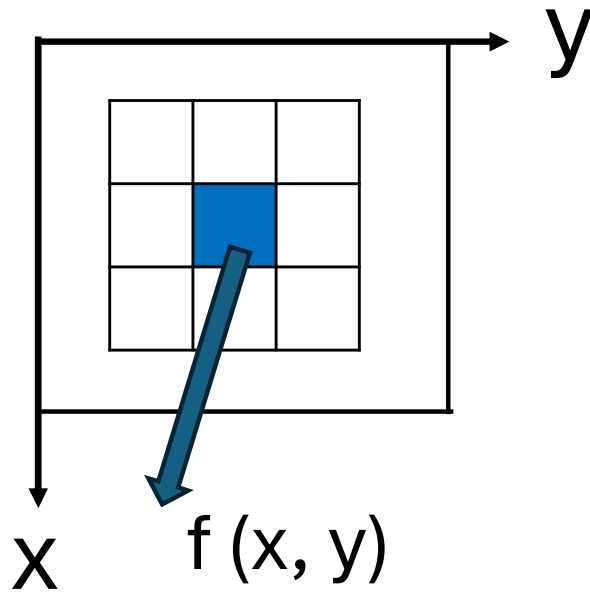
7×7

w1	w2	w3	w4	w5	w6	W7
w8	w9	w10	w11	w12	w13	w14
w15	w16	w17	w18	w19	w20	w21
w22	w23	w24	w25	w26	w27	w28
w29	w30	w31	w32	w33	w34	w35
w36	w37	w38	w39	w40	w41	w42
w43	w44	w45	w46	w47	w48	w49

5×5

n1	n2	n3	n4	n5
n6	n7	n8	n9	n10
n11	n12	n13	n14	n15
n16	n17	n18	n19	n20
n21	n22	n23	n24	n25

Spatial Filtering



	-1	0	+1
-1	$w(-1, -1)$	$w(-1, 0)$	$w(-1, 1)$
0	$w(0, -1)$	$w(0, 0)$	$w(0, 1)$
+1	$w(1, -1)$	$w(1, 0)$	$w(1, 1)$

	$y - 1$	y	$y + 1$
$x - 1$	$f(x - 1, y - 1)$	$f(x - 1, y)$	$f(x - 1, y + 1)$
x	$f(x, y - 1)$	$f(x, y)$	$f(x, y + 1)$
$x + 1$	$f(x + 1, y - 1)$	$f(x + 1, y)$	$f(x + 1, y + 1)$

Spatial Filtering

	-1	0	+1		y - 1	y	y + 1
-1	$w(-1,-1)$	$w(-1,0)$	$w(-1,1)$	x-1	$f(x-1, y-1)$	$f(x-1, y)$	$f(x-1, y+1)$
0	$w(0,-1)$	$w(0,0)$	$w(0,1)$	x	$f(x, y-1)$	$f(x, y)$	$f(x, y+1)$
+1	$w(1,-1)$	$w(1,0)$	$w(1,1)$	x+1	$f(x+1, y-1)$	$f(x+1, y)$	$f(x+1, y+1)$

$$R = f(x-1,y-1) \times w(-1,-1) + f(x-1,y) \times w(-1,0) + \dots + f(x+1,y+1) \times w(1,1)$$

In General Form

$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) \times f(x + s, y + t)$$

Spatial Filtering

	-1	0	+1
-1	$w(-1,-1)$	$w(-1,0)$	$w(-1,1)$
0	$w(0,-1)$	$w(0,0)$	$w(0,1)$
+1	$w(1,-1)$	$w(1,0)$	$w(1,1)$

	$y - 1$	y	$y + 1$
$x-1$	$f(x-1, y-1)$	$f(x-1, y)$	$f(x-1, y+1)$
x	$f(x, y-1)$	$f(x, y)$	$f(x, y+1)$
$x+1$	$f(x+1, y-1)$	$f(x+1, y)$	$f(x+1, y+1)$

In General Form
$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) \times f(x + s, y + t)$$

Spatial filtering modifies an image by replacing the value of each pixel by a function of the values of the pixel and its neighbours

Spatial Filtering

□ Spatial filtering modifies an image by replacing the value of each pixel by a function of the values of the pixel and its neighbours.

- If the operation performed on the image pixels is linear, then the filter is called as **Linear Spatial Filter**.

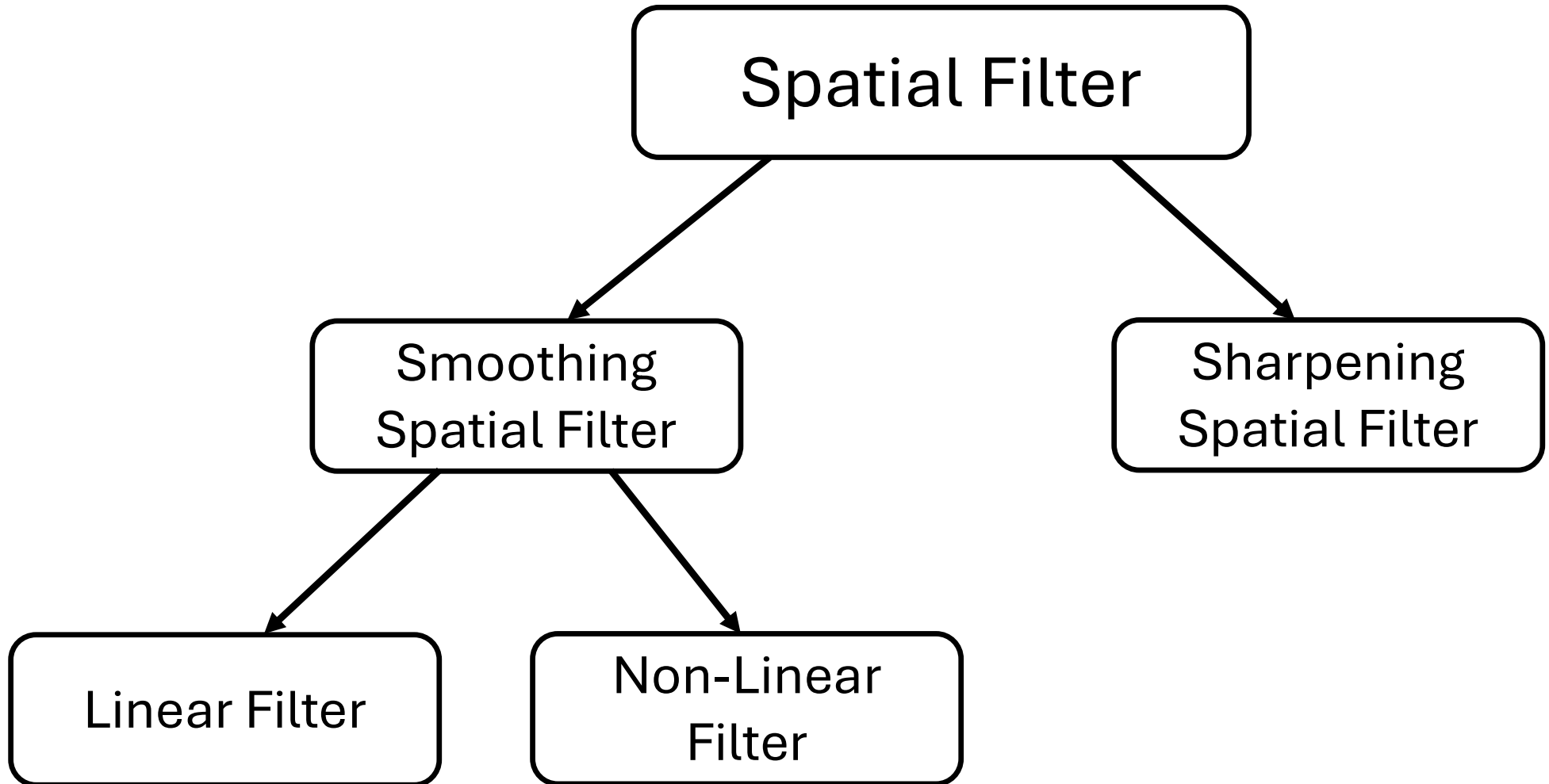
Example: Average filter

- Otherwise, the filter is a **Non-Linear Spatial Filter**. (Also known as **Order-Statistics Filter**)

Example: Median filter, Max filter, Min filter.

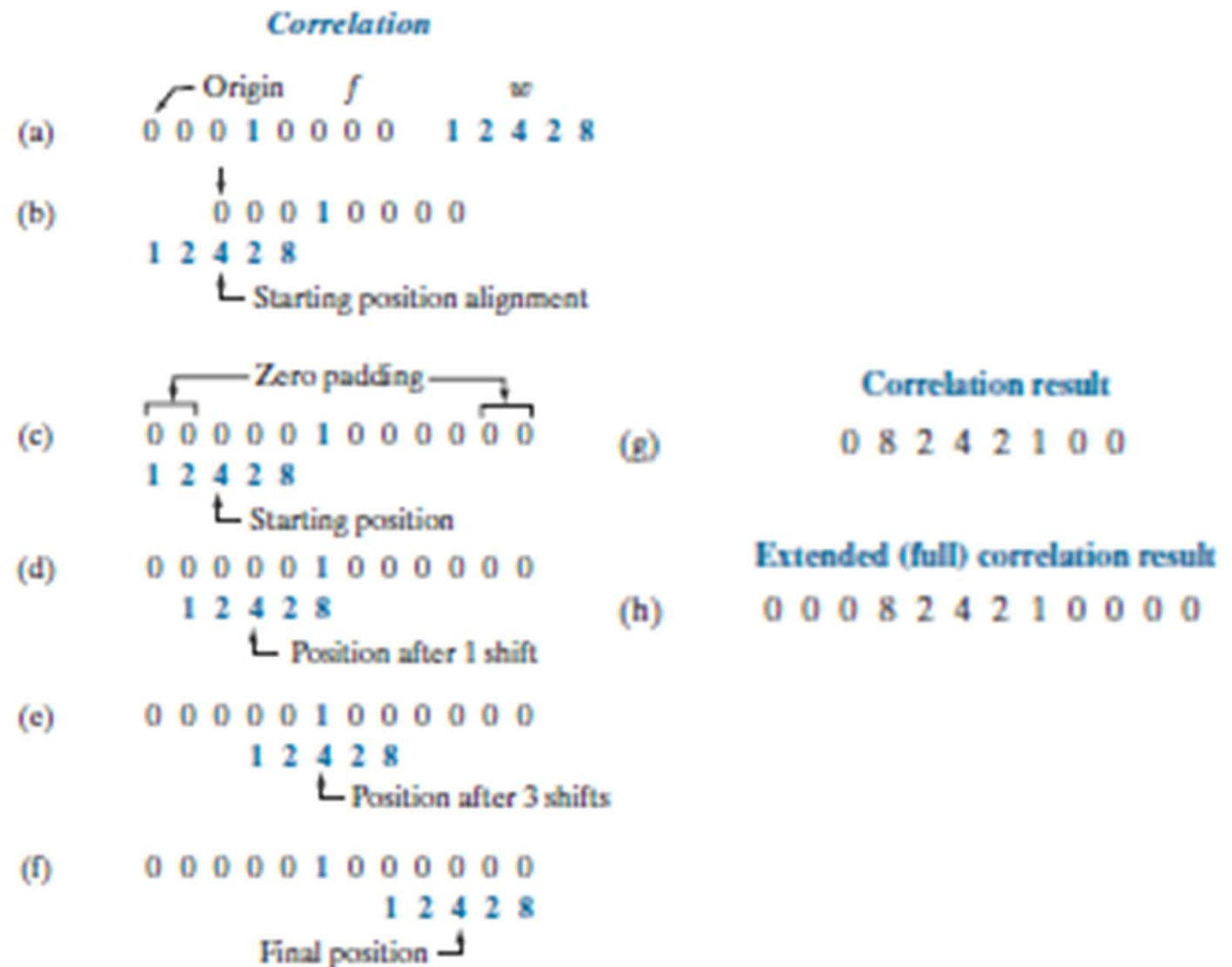
Spatial Filtering

❑ Classification of Spatial Filtering:



Correlation Operation

- Correlation:** Correlation consist of moving the center of the Kernel over an image and computing the sum of products at each locations.



One-Dimensional
(1-D)

Correlation Operation

X

0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---

W

1	2	3	2	8
---	---	---	---	---

0	0	0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---	---	---

1	2	3	2	8
---	---	---	---	---

Result = 0

0	0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---	---

1	2	3	2	8
---	---	---	---	---

Result = 8

0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---

1	2	3	2	8
---	---	---	---	---

Result = 2

0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---

1	2	3	2	8
---	---	---	---	---

Result = 3

0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---

1	2	3	2	8
---	---	---	---	---

Result = 2

0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---

Result = 1

1	2	3	2	8
---	---	---	---	---

0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---

0

1	2	3	2	8
---	---	---	---	---

Result = 0

Correlation Results

0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---

0 0

1	2	3	2	8
---	---	---	---	---

Result = 0

0	8	2	3	2	1	0	0
---	---	---	---	---	---	---	---

Correlation Operation

X								W				
0	0	0	1	0	0	0	0	1	2	3	2	8

Correlation Results

0	8	2	3	2	1	0	0
---	---	---	---	---	---	---	---

Full Correlation Results

0	0	0	8	2	3	2	1	0	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---

Convolution Operation

- Convolution:** The mechanism of spatial convolution are the same, except that the correlation Kernel is rotated by 180 degree.

Convolution

Origin f w rotated 180°
 0 0 0 1 0 0 0 0 8 2 4 2 1 (i)

0 0 0 1 0 0 0 0
 8 2 4 2 1 (j)

Starting position alignment

Zero padding
 0 0 0 0 0 1 0 0 0 0 0 0
 8 2 4 2 1 (k)

Starting position

0 0 0 0 0 1 0 0 0 0 0 0
 8 2 4 2 1 (l)

Position after 1 shift

0 0 0 0 0 1 0 0 0 0 0 0
 8 2 4 2 1 (m)

Position after 3 shifts

0 0 0 0 0 1 0 0 0 0 0 0
 8 2 4 2 1 (n)

Final position

Convolution result

0 1 2 4 2 8 0 0 (o)

Extended (full) convolution result

0 0 0 1 2 4 2 8 0 0 0 0 (p)

One-Dimensional
 (1-D)

Convolution Operation

X

0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---

W

1	2	3	2	8
---	---	---	---	---

w by 180 degree

8	2	3	2	1
---	---	---	---	---

0	0	0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---	---	---

8	2	3	2	1
---	---	---	---	---

Result = 0

0	0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---	---

8	2	3	2	1
---	---	---	---	---

Result = 1

0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---

8	2	3	2	1
---	---	---	---	---

Result = 2

0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---

8	2	3	2	1
---	---	---	---	---

Result = 3

0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---

8	2	3	2	1
---	---	---	---	---

Result = 2

0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---

Result = 8

8	2	3	2	1
---	---	---	---	---

0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---

8	2	3	2	1
---	---	---	---	---

Result = 0

Convolution Results

0	0	0	1	0	0	0	0	0	0
---	---	---	---	---	---	---	---	---	---

8	2	3	2	1
---	---	---	---	---

Result = 0

0	1	2	3	2	8	0	0
---	---	---	---	---	---	---	---

Convolution Operation

X								W				
0	0	0	1	0	0	0	0	1	2	3	2	8

Convolution Results

0	1	2	3	2	8	0	0
---	---	---	---	---	---	---	---

Full Convolution Results

0	0	0	1	2	3	2	8	0	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---

Correlation and Convolution Operation

X

0	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---

W

1	2	3	2	8
---	---	---	---	---

Correlation Results

0	8	2	3	2	1	0	0
---	---	---	---	---	---	---	---

Convolution Results

0	1	2	3	2	8	0	0
---	---	---	---	---	---	---	---

Numerical Examples

Ques1: How is convolution and Correlation used for linear spatial filtering? Perform each of them on 1D function $f = 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0$ and a filter $w = 1 \ 2 \ 3 \ 2 \ 8$

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Initial position for w									Full correlation result									Cropped correlation result				
<u>1</u>	<u>2</u>	<u>3</u>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<u>4</u>	<u>5</u>	<u>6</u>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	8	7	0
<u>7</u>	<u>8</u>	<u>9</u>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	5	4	0
0	0	0	0	0	0	0	0	0	0	0	0	9	8	7	0	0	0	0	3	2	1	0
0	0	0	0	1	0	0	0	0	0	0	0	6	5	4	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	3	2	1	0	0	0					
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					

Two-Dimensional (2-D)

Numerical Examples

Ques1: Use the following 3×3 mask to perform the convolution process on the shaded pixels in the 5×5 image below. Write the filtered image.

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0	$1/6$	0
$1/6$	$1/3$	$1/6$
0	$1/6$	0

3×3 Mask

30	40	50	70	90
40	50	80	60	100
35	255	70	0	120
30	45	80	100	130
40	50	90	125	140

5×5 Image

Correlation and Convolution

- Mathematical Expression of Correlation Operation in two dimensional:

$$w(x, y) \star f(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)$$

- Mathematical Expression of Convolution Operation in two dimensional:

$$w(x, y) \star f(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x - s, y - t)$$

where,

$$a = (m-1)/2$$

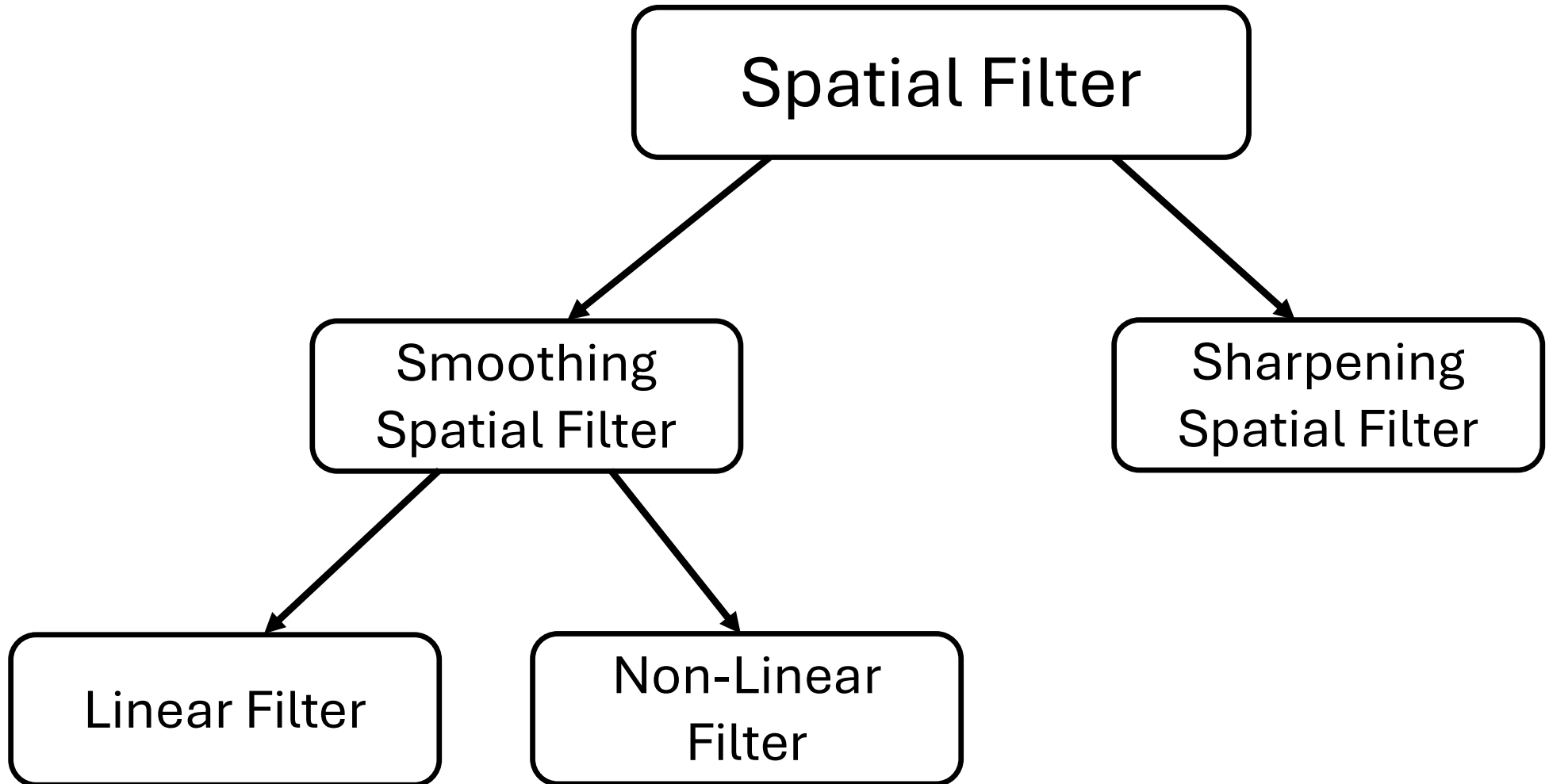
M, N: be the dimension of the image

$$b = (n-1)/2$$

m, n: be the dimension of the mask

Spatial Filtering

❑ Classification of Spatial Filtering:



Smoothing Spatial Filter

❑ Smoothing Filters:

- To remove the noise present in the image
- To blur the image
- To extract the object, present in the image
- To remove the small object, present in the image
- To bridge the gap between lines in an image.

❑ Types of Smoothing Spatial Filters:

- Linear Filter
- Non-Linear or Order-Statistics Spatial Filter

Smoothing (Linear) Filter

Smoothing Linear Spatial Filters:

- To remove the noise in an image
- To Blur the image
- Example: Averaging Filter

Averaging Filter: Determine the average considering its neighbours and assigning it to its center pixel.

5×5

3×3

$$\left(\frac{1}{9}\right) \times \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

$$R = \frac{1}{9} \sum_{i=1}^9 W_i Z_i$$

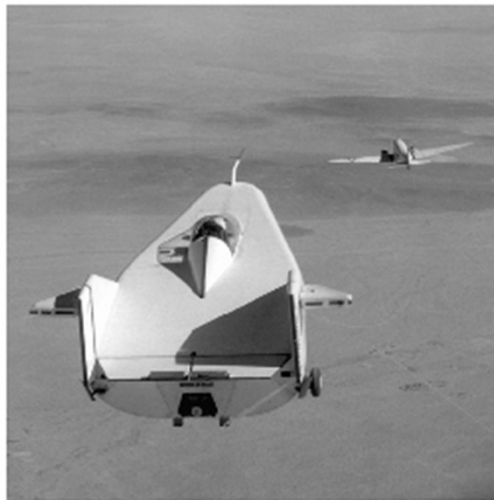
$$\left(\frac{1}{25}\right) \times$$

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

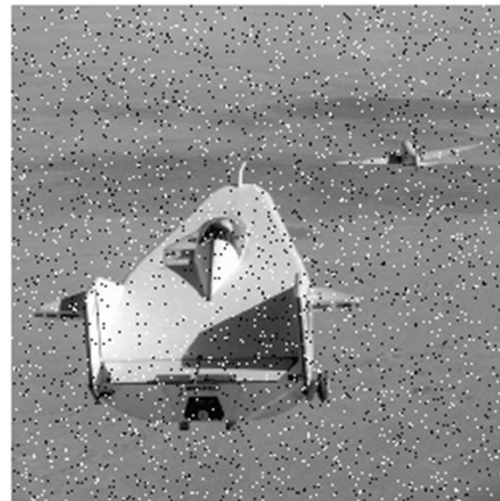
$$R = \frac{1}{25} \sum_{i=1}^{25} W_i Z_i$$

Smoothing (Linear) Filter

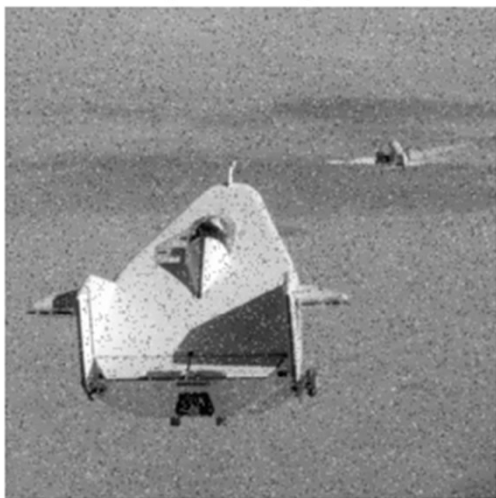
Original Image A



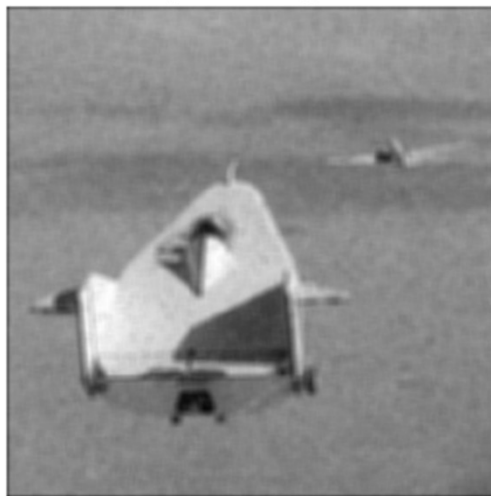
Noisy Image B: salt & pepper



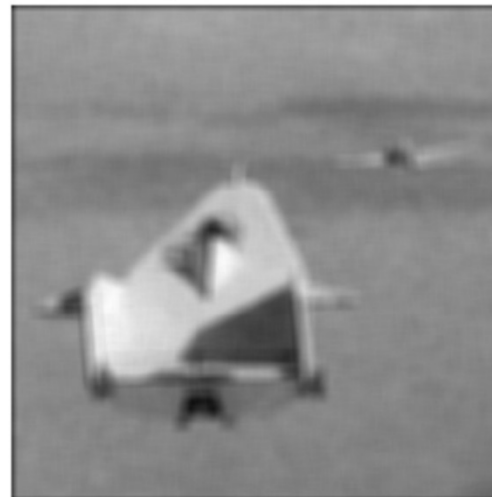
Output Image C3



Output Image C9



Output Image C15



Output Image C35



Smoothing (Linear) Filter

❑ Weighted-Average Filter:

$$\left(\frac{1}{16}\right) \times \begin{array}{|c|c|c|} \hline & 3 \times 3 & \\ \hline 1 & 2 & 1 \\ \hline 2 & 4 & 2 \\ \hline 1 & 2 & 1 \\ \hline \end{array}$$

$$R = \frac{1}{9} \sum_{i=1}^9 W_i Z_i$$

❑ General Mathematical Expression for Average Filter:

$$g(x, y) = \frac{\sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)}{\sum_{s=-a}^a \sum_{t=-b}^b w(s, t)}$$

Smoothing (Non-Linear) Filter

□ Smoothing Non-Linear Filters:

- Used to remove the noise in the image
- Used in salt and pepper noise
- **Examples:** Median filter, Max filter and Min filter

- **Median Filter:** Median Value is computed from its neighbourhood and that median value is assigned to center pixels.

$$R = \text{median}\{z_k \mid k = 1, 2, \dots, n \times n\}$$

- **Max Filter:** Maximum Value is computed from its neighbourhood and that maximum value is assigned to center pixels.

$$R = \max\{z_k \mid k = 1, 2, \dots, n \times n\}$$

- **Min Filter:** Minimum Value is computed from its neighbourhood and that minimum value is assigned to center pixels.

$$R = \min\{z_k \mid k = 1, 2, \dots, n \times n\}$$

Numerical Examples

Ques2: Consider the following image

0	1	0	2	7
2	1	6	1	0
5	6	7	6	3
1	1	6	1	5
5	4	2	2	5

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2021-22**

What will be the new value of the pixel (2,2) if smoothing is done using

- (a) Mean Filter
- (b) Weight average filter (Assign weight as 3, 2, and 1)
- (c) Median Filter
- (d) Min Filter
- (e) Max Filter

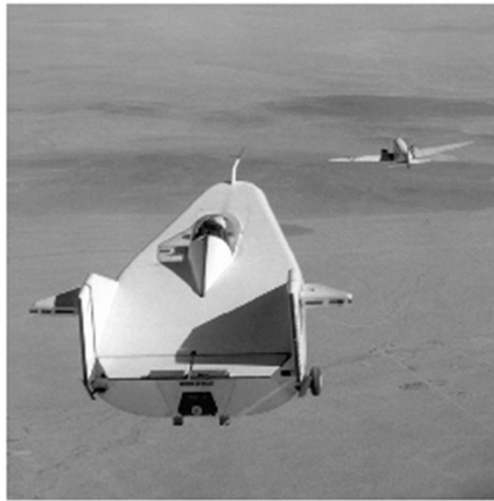
Numerical Examples

Ques 1: For the following digital image, Determine the output gray intensity level of pixel P by using (a) 3×3 standard average filter, and (b) 3×3 Weight average filter

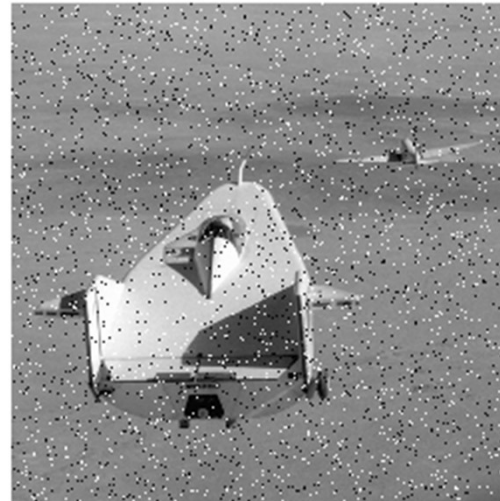
110	120	90	130
91	P 94	98	200
90	91	99	100
82	96	85	90

Analysing Average Filter

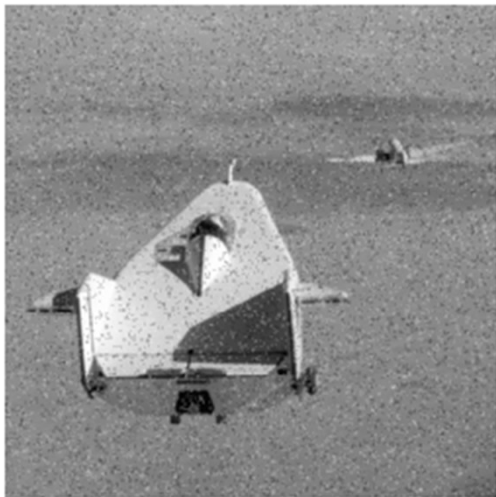
Original Image A



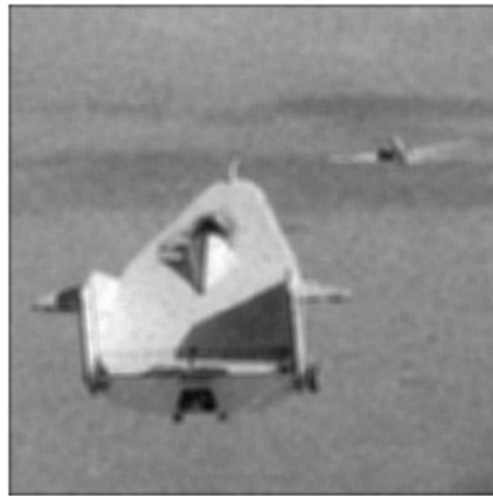
Noisy Image B: salt & pepper



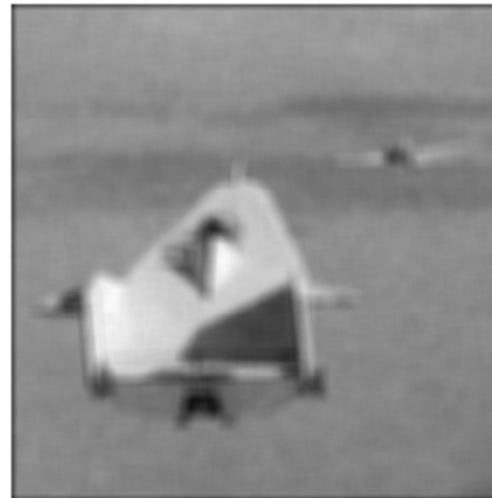
Output Image C3



Output Image C9



Output Image C15

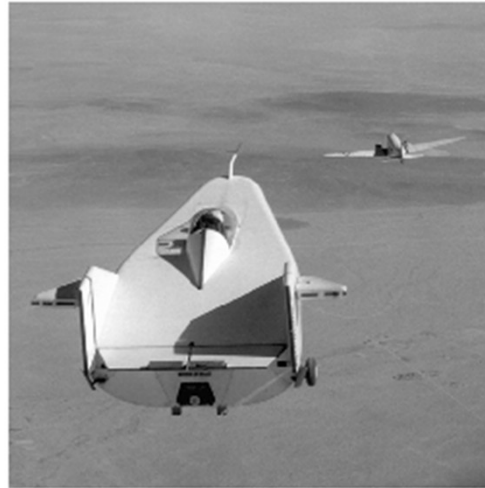


Output Image C35

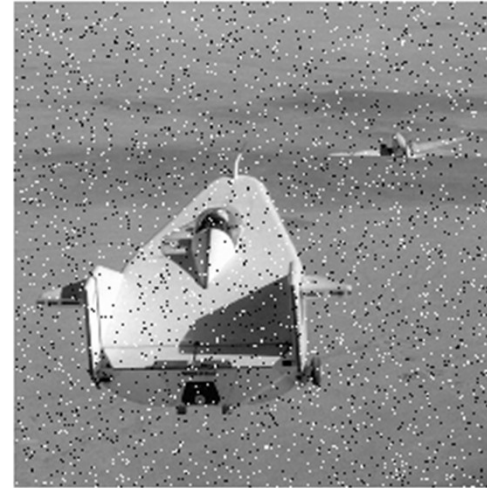


Analysing Min Filter

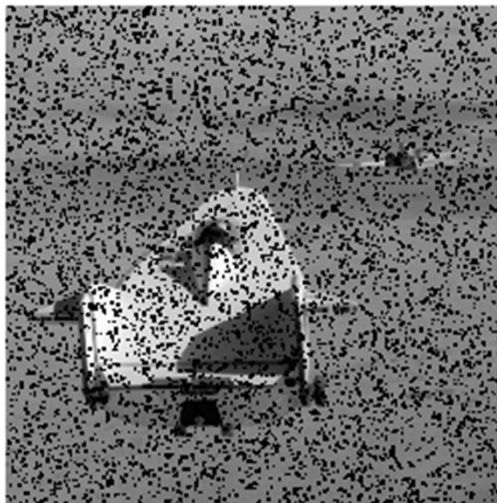
Original Image A



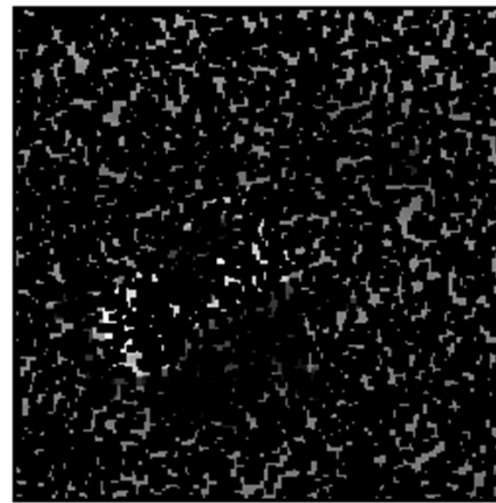
Noisy Image B: salt and Pepper



Output Min 3*3 Image



Output Min 9*9 Image

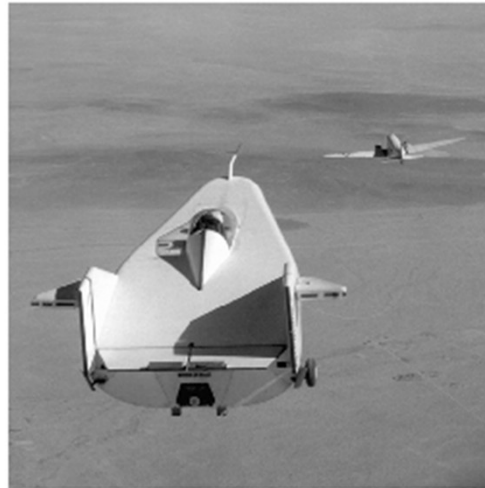


Output Min 15*15 Image

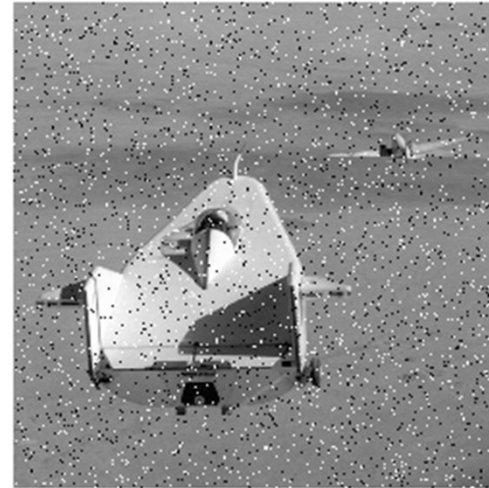


Analysing Max Filter

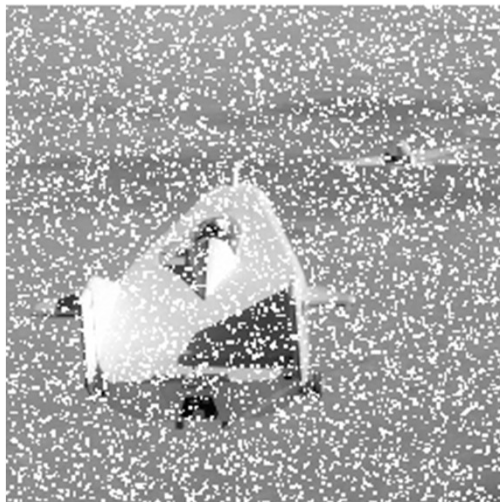
Original Image A



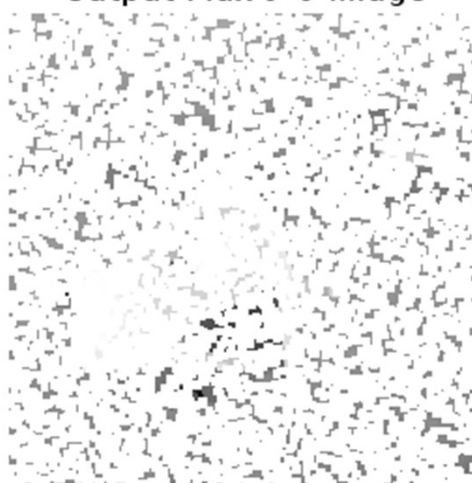
Noisy Image B: salt and Pepper



Output Max 3*3 Image



Output Max 9*9 Image

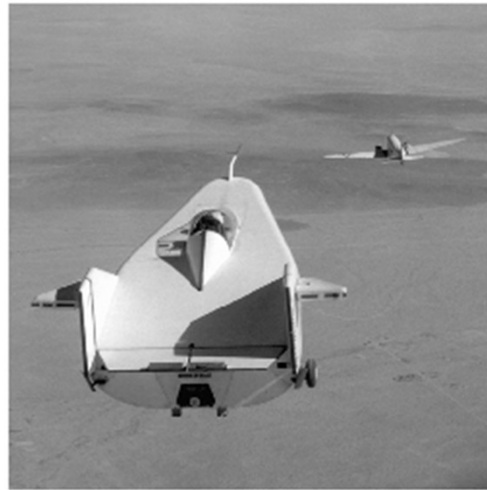


Output Max 15*15 Image

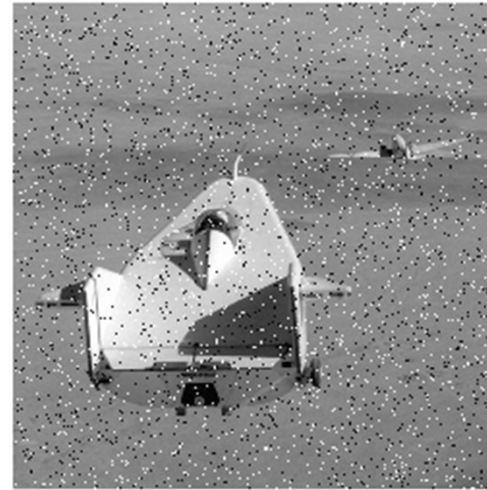


Analysing Med Filter

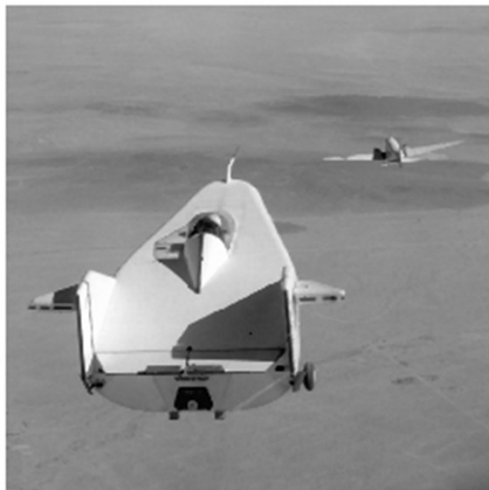
Original Image A



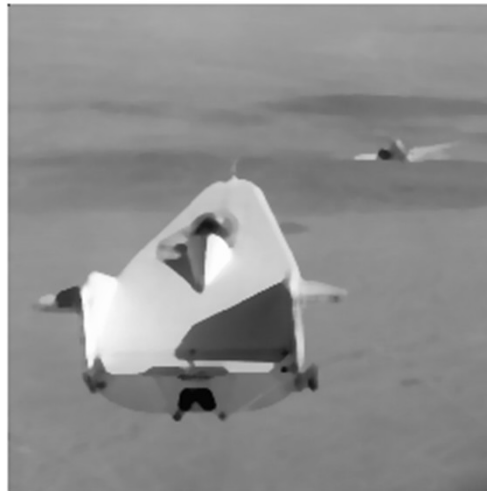
Noisy Image B: salt and Pepper



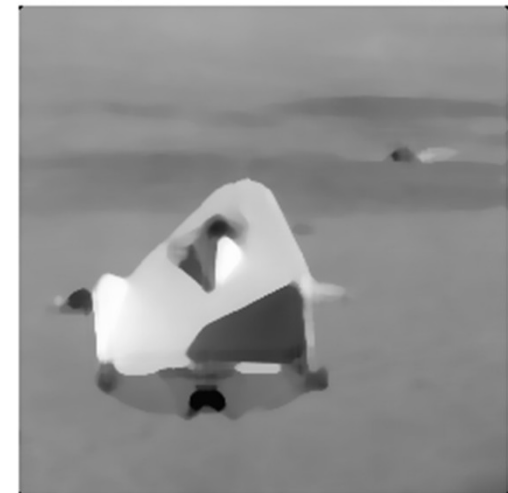
Output Med 3*3 Image



Output Med 9*9 Image



Output Med 15*15 Image



Sharpening Spatial Filter

Sharpening Spatial Filter

- Used to highlight the transition in intensities.
- Perform reverse operation on smoothing spatial filter.
- In smoothing we perform averaging operation which can be considered as pixel integration; whereas in sharpening spatial filter we perform pixel differentiation.
- Used to highlight the fine details in an image
- It enhances the blurred image
- It enhances the edges.

Sharpening Spatial Filter

□ Applications:

- Used in electronic printing
- Medical imaging
- Industrial inspection
- Autonomous guidance military system

□ Spatial differentiation can be performed by using

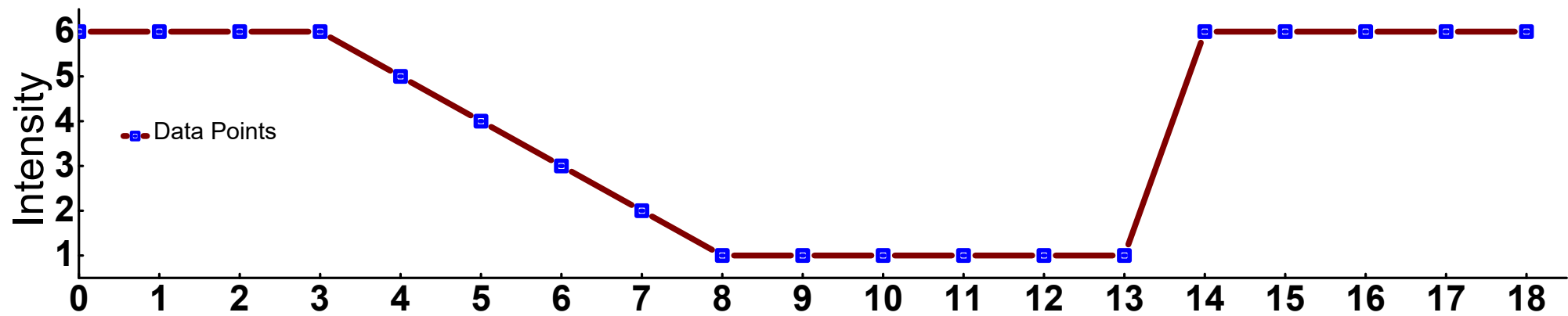
- First order derivative
- Second order derivative

6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

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[illegible]

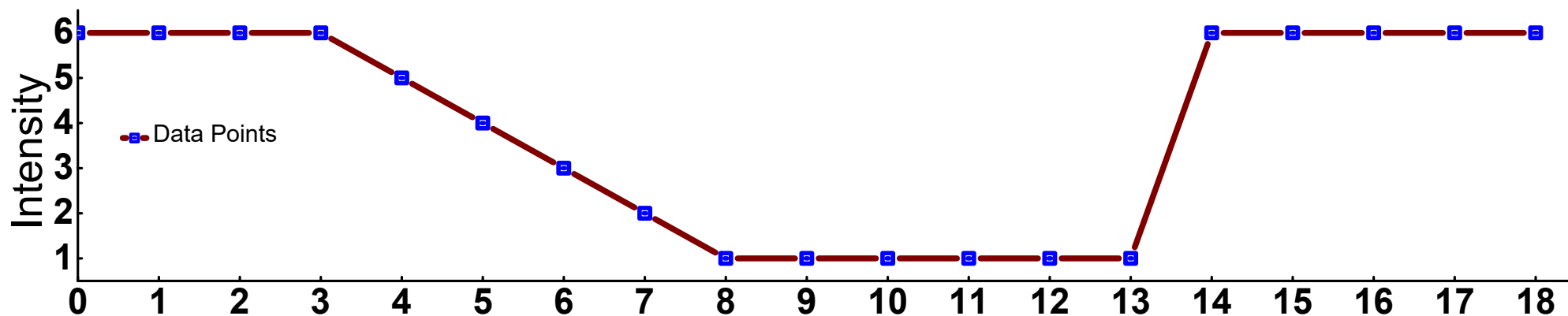
First Order Derivative



Mathematical Equation of 1st order Derivative: $\frac{\partial f}{\partial x} = f(x + 1) - f(x)$

x	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
f(x)	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6
$\frac{\partial f}{\partial x}$	0	0	0	-1	-1	-1	-1	-1	0	0	0	0	0	5	0	0	0	0	

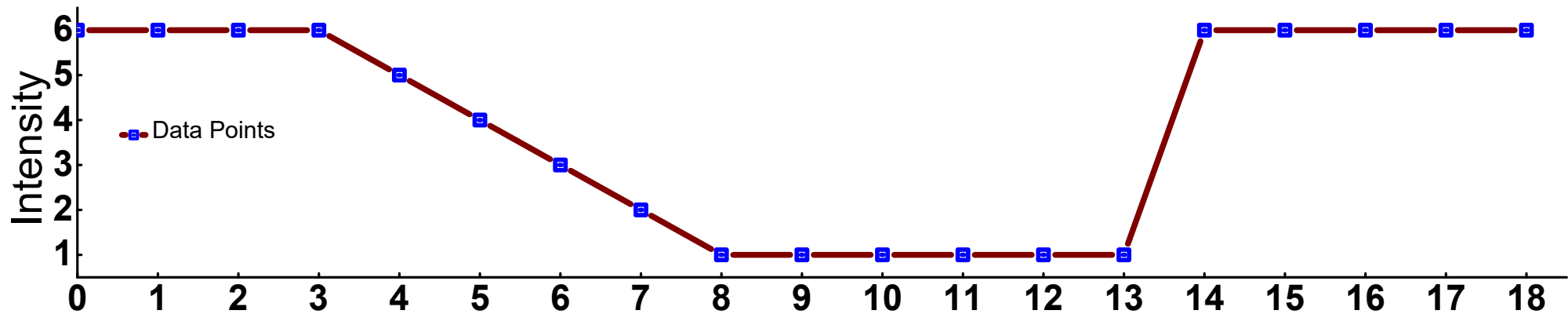
First Order Derivative



x	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$f(x)$	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6
$\frac{\partial f}{\partial x}$	0	0	0	-1	-1	-1	-1	-1	0	0	0	0	0	5	0	0	0	0	

- **For constant Intensity:** First order derivative $\frac{\partial f}{\partial x} = 0$
- **Onset of ramp/steps:** First order derivative $\frac{\partial f}{\partial x} \neq 0$

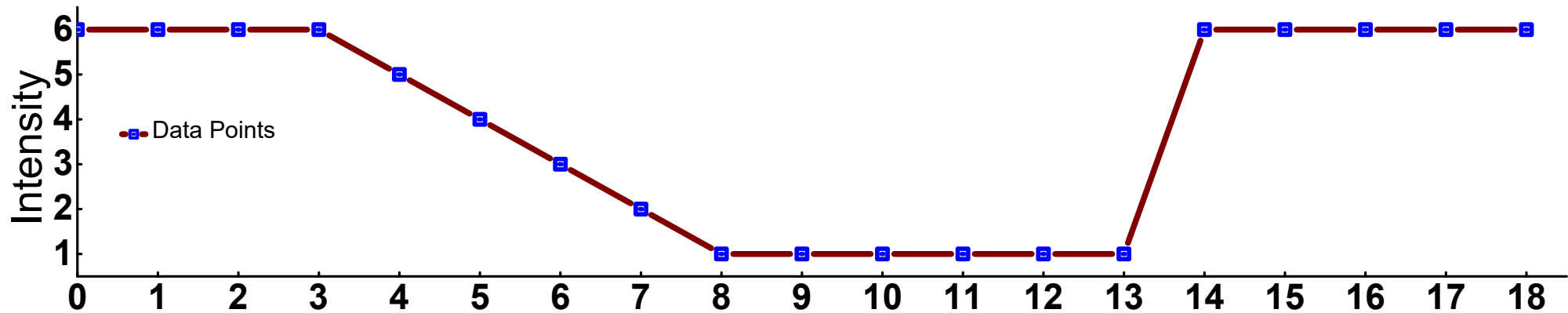
Second Order Derivative



Equation of 2nd order Derivative: $\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$

x	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$f(x)$	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6
$\frac{\partial^2 f}{\partial x^2}$		0	0	-1	0	0	0	0	1	0	0	0	0	5	-5	0	0	0	

Second Order Derivative



- For constant Intensity: $\frac{\partial^2 f}{\partial x^2} = \text{zero}$
- Onset and end of the ramp: $\frac{\partial^2 f}{\partial x^2} = \text{non-zero}$
- Onset and end of the step: $\frac{\partial^2 f}{\partial x^2} = \text{non-zero}$
- Constant slope ramp: $\frac{\partial^2 f}{\partial x^2} = \text{zero}$

Second Order Derivative for Images

2nd Order Derivative for Images

- The second order derivative for 2D $f(x, y)$ is given by:

$$\nabla^2 f(x, y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

- In x-direction:

$$\frac{\partial^2 f}{\partial x^2} = f(x + 1, y) + f(x - 1, y) - 2f(x, y)$$

- In y-direction:

$$\frac{\partial^2 f}{\partial y^2} = f(x, y + 1) + f(x, y - 1) - 2f(x, y)$$

- So,

$$\nabla^2 f(x, y) = f(x + 1, y) + f(x - 1, y) + f(x, y + 1) + f(x, y - 1) - 4f(x, y)$$

2nd Order Derivative for Images

Laplacian Formulation Equation:

$$\nabla^2 f(x, y) = f(x + 1, y) + f(x - 1, y) + f(x, y + 1) + f(x, y - 1) - 4f(x, y)$$

	$y - 1$	y	$y + 1$
$x - 1$	$f(x - 1, y - 1)$	$f(x - 1, y)$	$f(x - 1, y + 1)$
x	$f(x, y - 1)$	$f(x, y)$	$f(x, y + 1)$
$x + 1$	$f(x + 1, y - 1)$	$f(x + 1, y)$	$f(x + 1, y + 1)$

0	1	0
1	-4	1
0	1	0

□ For Sharpening Application:

- Necessary condition for Mask: Sum of all the co-efficient must be equal to zero.

2nd Order Derivative for Images

- Considering the diagonal elements also:

1	1	1
1	-8	1
1	1	1

- General expression to find out sharpened image:

$$g(x, y) = f(x, y) + c[\nabla^2 f(x, y)]$$

Where, C is constant.

if $C = -1$, Mask used are

0	1	0
1	-4	1
0	1	0

1	1	1
1	-8	1
1	1	1

if $C = 1$, Mask used are

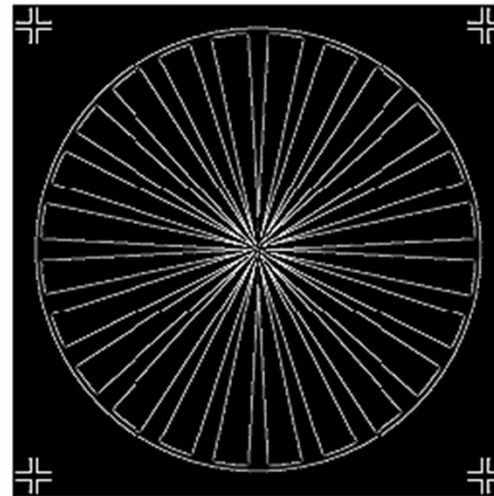
0	-1	0
-1	4	-1
0	-1	0

-1	-1	-1
-1	8	-1
-1	-1	-1

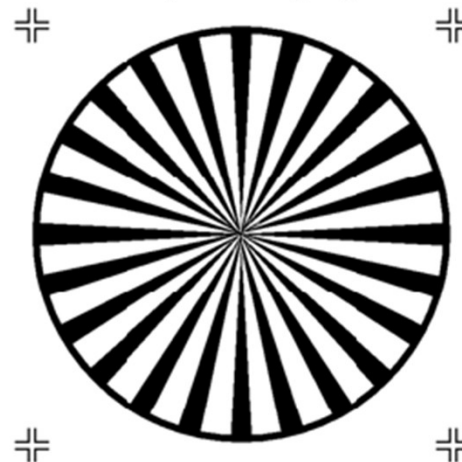
2nd Order Derivative for Images

0	1	0
1	-4	1
0	1	0

Laplacian output Image out4 using h4

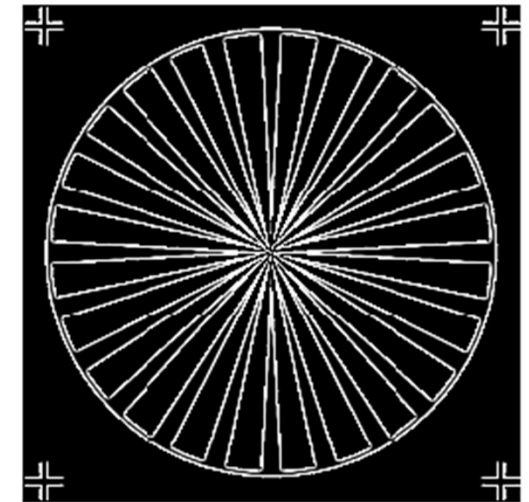


Output Image g4

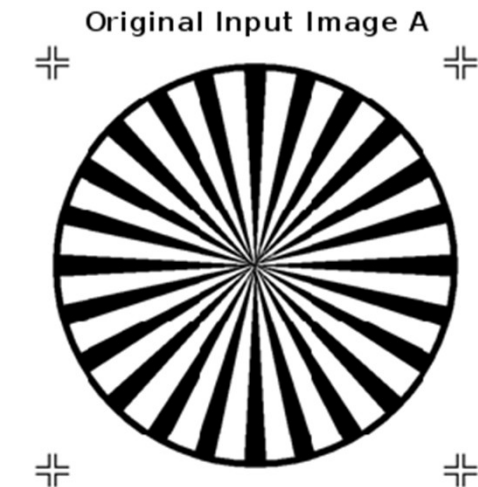
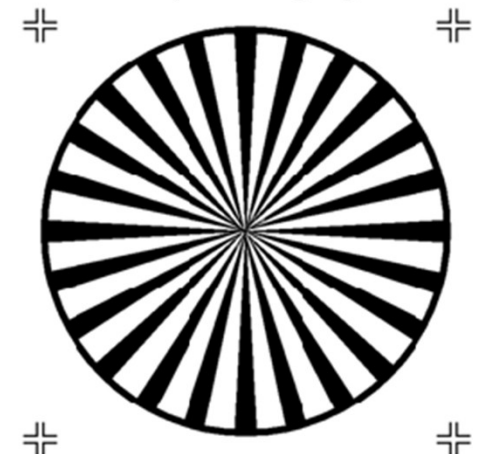


1	1	1
1	-8	1
1	1	1

Laplacian output Image out8 using h8



Output Image g8



$$g(x, y) = f(x, y) + c[\nabla^2 f(x, y)]$$

$$C = -1$$

2nd Order Derivative for Images

a b
 c d

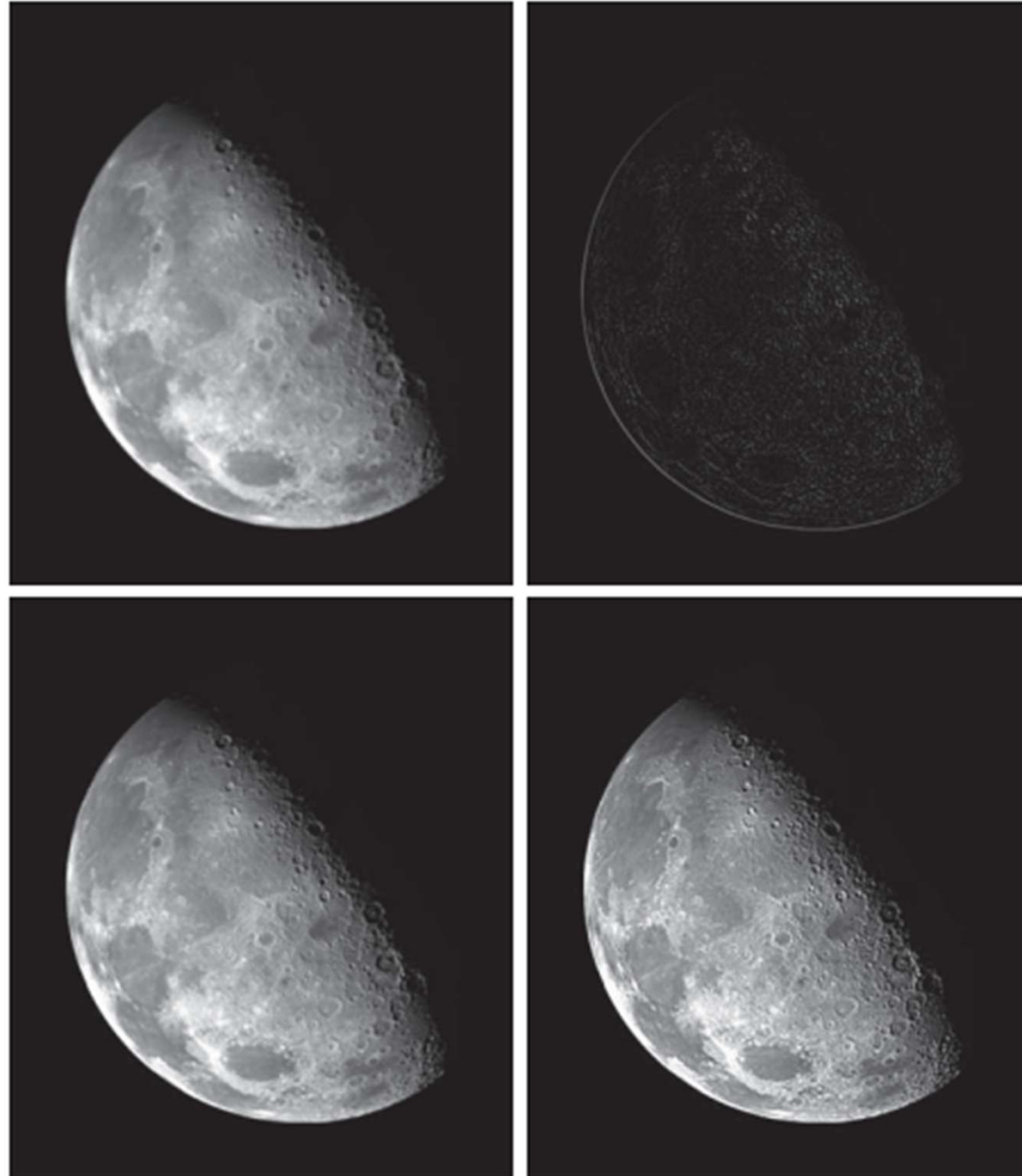
FIGURE 3.46

(a) Blurred image of the North Pole of the moon.

(b) Laplacian image obtained using the kernel in Fig. 3.45(a).

(c) Image sharpened using Eq. (3-54) with $c = -1$.

(d) Image sharpened using the same procedure, but with the kernel in Fig. 3.45(b). (Original image courtesy of NASA.)



Numerical Examples

Ques 8: Compute the convolution of the Laplacian Kernel L_8 with the image given below. Use border values to extend the image.

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100	100	150	150
100	100	150	150
100	100	150	150
100	100	150	150

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[illegible]

Numerical Examples

Ques: What are Sharpening spatial filter? Discuss the response of first and second order derivative. What is the Laplacian mask for image sharpening?

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Numerical Examples

Ques 4: if $X = \{2 \ 3 \ 4 \ 3 \ 4 \ 5 \ 6\}$, and $W = \{1 \ \underline{1} \ 1\}$. What would be the result of median filtering? Use zero padding if required.

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Numerical Examples

Ques5: Consider an image shown in Fig.5 (alias I6). Suppose we apply a 1×3 average filter on an image I6 to get the image I7. Again we apply the same filter on image I7 to get the image I8.

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2	5	6	6	7	4	3
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Find the final value of the center pixel.

Numerical Examples

Ques5: A mask is defined with the following mathematical expression:

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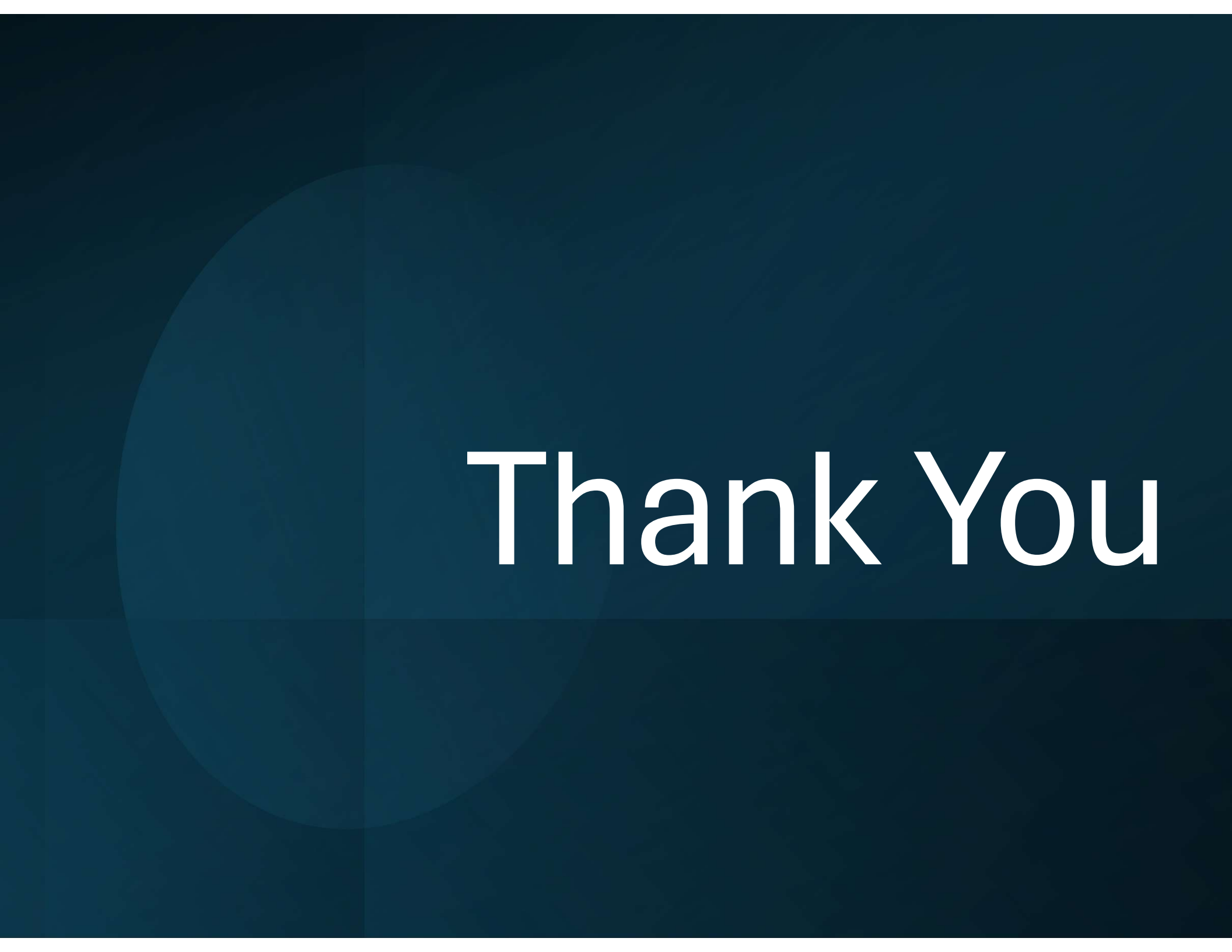
$$f(x-2,y) + 2f(x-1, y) + 2f(x+1, y) + f(x+2, y) + f(x, y-2) + 2f(x, y-1) + 2f(x, y+1) + f(x, y+2) - 12f(x, y).$$

Draw this mask.

9	9	9	8	7	6	5	4	2	2	2	2	8	8	6	6	6	6	6
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Your answer should be in tabular form having 3 rows as shown below. First row should show original signal, second row should have values for the first-order derivative and the third row should have values for the second order derivative

[illegible]



Thank You